10.1 Solution: (a) The differential scattering cross section for a unit vector $\mathbf{n}$ and one associated polarization $\boldsymbol{\epsilon}$ is

$$\begin{aligned}
\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0;\mathbf{n},\boldsymbol{\epsilon}) &= k^4a^6\left|\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0-\frac{1}{2}(\mathbf{n}\times\boldsymbol{\epsilon}^*)\cdot(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)\right|^2\\
&= k^4a^6\left|\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)(\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0)+\frac{1}{2}(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)(\boldsymbol{\epsilon}^*\cdot\mathbf{n}_0)\right|^2\\
&= k^4a^6\left|\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)(\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0)+\frac{1}{2}(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)(\boldsymbol{\epsilon}^*\cdot\mathbf{n}_0)\right|^2\\
&= k^4a^6\Bigg[\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)^2|\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0|^2+\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)(\boldsymbol{\epsilon}^*\cdot\boldsymbol{\epsilon}_0)(\boldsymbol{\epsilon}^*\cdot\mathbf{n}_0)\\
&\qquad+\frac{1}{4}|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2|\boldsymbol{\epsilon}^*\cdot\mathbf{n}_0|^2\Bigg].
\end{aligned}$$

The scattering cross section, summed over outgoing polarizations, is

$$\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0,\mathbf{n})=\sum_\lambda\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0;\mathbf{n},\boldsymbol{\epsilon}_\lambda).$$

To perform the above summation, the following identity will be useful,

$$\sum_{\lambda=1,2}\epsilon^*_{\lambda,i}\epsilon_{\lambda,j}=\delta_{ij}-n_in_j.$$

Then,

$$\sum_{\lambda=1,2}(\boldsymbol{\epsilon}^*_\lambda\cdot\mathbf{a})(\boldsymbol{\epsilon}_\lambda\cdot\mathbf{b})=\mathbf{a}\cdot\mathbf{b}-(\mathbf{n}\cdot\mathbf{a})(\mathbf{n}\cdot\mathbf{b}).$$

By the above relation, we now have

$$\begin{aligned}
\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0,\mathbf{n}) &= k^4a^6\Bigg[\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)^2\left(1-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\right)\\
&\qquad+\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)\left[\mathbf{n}_0\cdot\boldsymbol{\epsilon}_0-(\mathbf{n}\cdot\mathbf{n}_0)(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)\right]\\
&\qquad+\frac{1}{4}|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\left(1-|\mathbf{n}\cdot\mathbf{n}_0|^2\right)\Bigg].
\end{aligned}$$

This can be slightly simplied by noticing that $\mathbf{n}_0\cdot\boldsymbol{\epsilon}_0=0$,

$$\begin{aligned}
\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0,\mathbf{n}) &= k^4a^6\Bigg[\left(1-\mathbf{n}\cdot\mathbf{n}_0+\frac{1}{4}|\mathbf{n}\cdot\mathbf{n}_0|^2\right)\left(1-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\right)\\
&\qquad+\left(1-\frac{1}{2}(\mathbf{n}\cdot\mathbf{n}_0)\right)(\mathbf{n}\cdot\mathbf{n}_0)|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\\
&\qquad+\frac{1}{4}|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\left(1-|\mathbf{n}\cdot\mathbf{n}_0|^2\right)\Bigg]
\end{aligned}$$

$$
\begin{aligned}
&= k^4a^6\Big[1-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2-\mathbf{n}\cdot\mathbf{n}_0+(\mathbf{n}\cdot\mathbf{n}_0)(\mathbf{n}\cdot\boldsymbol{\epsilon}_0)^2+\frac{1}{4}|\mathbf{n}\cdot\mathbf{n}_0|^2-\frac{1}{4}|\mathbf{n}\cdot\mathbf{n}_0|^2|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\\
&\qquad -|\mathbf{n}\cdot\mathbf{n}_0||\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2+\frac{1}{2}|\mathbf{n}\cdot\mathbf{n}_0|^2|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\\
&\qquad +\frac{1}{4}|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2-\frac{1}{4}|\mathbf{n}\cdot\mathbf{n}_0|^2|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\Big]\\
&= k^4a^6\left[1-\mathbf{n}\cdot\mathbf{n}_0-\frac{3}{4}|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2+\frac{1}{4}|\mathbf{n}\cdot\mathbf{n}_0|^2\right].
\end{aligned}
$$

This can be further simplified by the following identity,

$$
|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2+|\mathbf{n}\cdot\mathbf{n}_0|^2+|\mathbf{n}\cdot(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)|^2=1,
$$

as $\mathbf{n}_0$, $\boldsymbol{\epsilon}_0$, and $\mathbf{n}_0\times\boldsymbol{\epsilon}_0$ form a complete base. Then,

$$
\begin{aligned}
\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0,\mathbf{n}) &= k^4a^6\left[1-\mathbf{n}\cdot\mathbf{n}_0-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2+\frac{1}{4}\left(|\mathbf{n}\cdot\mathbf{n}_0|^2+|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2\right)\right]\\
&= k^4a^6\left[1-\mathbf{n}\cdot\mathbf{n}_0-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2+\frac{1}{4}\left(1-|\mathbf{n}\cdot(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)|^2\right)\right]\\
&= k^4a^6\left[\frac{5}{4}-|\mathbf{n}\cdot\boldsymbol{\epsilon}_0|^2-\frac{1}{4}|\mathbf{n}\cdot(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)|^2-\mathbf{n}\cdot\mathbf{n}_0\right].
\end{aligned}
$$

(b) For linearly polarized incident radiations, we can choose $\mathbf{n}_0=(0,0,1)$, $\mathbf{n}=(\sin\theta\cos\phi,\sin\theta\sin\phi,\cos\theta)$, and $\boldsymbol{\epsilon}_0=(1,0,0)$. Then,

$$
\mathbf{n}\cdot\mathbf{n}_0=\cos\theta,\quad \mathbf{n}\cdot\boldsymbol{\epsilon}_0=\sin\theta\cos\phi,\quad \mathbf{n}\cdot(\mathbf{n}_0\times\boldsymbol{\epsilon}_0)=\sin\theta\sin\phi,
$$

and the scattering cross section is

$$
\begin{aligned}
\frac{d\sigma}{d\Omega}(\mathbf{n}_0,\boldsymbol{\epsilon}_0,\mathbf{n}) &= k^4a^6\left[\frac{5}{4}-\sin^2\theta\cos^2\phi-\frac{1}{4}\sin^2\theta\sin^2\phi-\cos\theta\right]\\
&= k^4a^6\left[\frac{5}{4}-\frac{1}{2}\sin^2\theta(\cos 2\phi+1)-\frac{1}{8}\sin^2\theta(1-\cos 2\phi)-\cos\theta\right]\\
&= k^4a^6\left[\frac{5}{4}-\frac{5}{8}\sin^2\theta-\frac{3}{8}\sin^2\theta\cos 2\phi-\cos\theta\right]\\
&= k^4a^6\left[\frac{5}{8}(1+\cos^2\theta)-\cos\theta-\frac{3}{8}\sin^2\theta\cos 2\phi\right].
\end{aligned}
$$

(c) From part (b), we have

$$
\frac{\dfrac{d\sigma}{d\Omega}\left(\theta=\dfrac{\pi}{2},\phi=0\right)}{\dfrac{d\sigma}{d\Omega}\left(\theta=\dfrac{\pi}{2},\phi=\dfrac{\pi}{2}\right)}=\frac{\dfrac{5}{8}-\dfrac{3}{8}}{\dfrac{5}{8}+\dfrac{3}{8}}=\frac{1}{4}.
$$

For $\theta=\pi/2$, $\phi=0$, the incident electric field will induce electric current in the polarization direction. However, electric dipole does not radiate in its dipole moment direction. Therefore, the radiation in the polarization direction is only due to the magnetic dipole. On the other hand, the magnetic dipole moment does not contribute perpendicular to this polarization. For $\theta=\frac{\pi}{2}$, $\phi=\pi/2$, we are only getting electric dipole radiation. From Eq. (10.14), we can see that magnetic dipole radiation strength is only a quarter of that of the electric dipole.